CRIPCOCODE ARTIFICIAL INTELLIGENCE PRIVATE LIMITED

CIN: U62091WB2025PTC282327
Address: 74/2 Incinerator Road, Lp 81/5/1/1, Dum Dum Cant, Nagerbazar, Kolkata, Kolkata, West Bengal, India, 700028

School Management System - Full Project Overview

Project Name: School Management System (Web + Mobile)

Company: Cripcocode Technologies Pvt Ltd

---

1. Project Overview

The School Management System is a complete digital ERP solution designed to manage and automate all academic and administrative operations of a school.

This system includes a web-based Admin CRM and mobile applications for Teachers, Students, and Parents.

The backend is developed using Node.js and MongoDB, and the mobile application is built using Flutter.

The system improves efficiency, reduces manual paperwork, ensures real-time data access, and enhances communication between school, teachers, students, and parents.

---

2. Project Objectives

- Automate school administrative operations

- Digital student admission and records management

- Teacher and staff management

- Attendance tracking for students and teachers

- Online fee collection and receipt generation

- Academic management including exams and results

- Real-time notifications and communication

- Mobile access for parents, students, and teachers

---

3. User Panels

1. Admin Panel (Web CRM)

2. Teacher Panel (Web + Mobile)

3. Student Panel (Mobile)

4. Parent Panel (Mobile)

5. Accountant Panel (Web)

---

4. Core Modules

- Student Management

- Teacher & Staff Management

- Class & Subject Management

- Attendance Management

- Fees & Finance Management

- Examination & Results

- Homework & Notices

- Communication & Notifications

- Reports & Analytics

---

5. Technology Stack

Backend:

- Node.js

- Express.js

- MongoDB

- Mongoose ORM

- JWT Authentication

Frontend Admin Panel:

- React.js / Next.js

- Tailwind CSS

Mobile Application:

- Flutter (Android & iOS)

Hosting:

- VPS / AWS / DigitalOcean

- MongoDB Atlas

Other Integrations:

- Payment Gateway (Razorpay / Stripe)

- Firebase Push Notifications

- SMS & Email APIs

---

6. Database Design (MongoDB Collections)

Users:

- name

- email

- password

- role

Students:

- studentId

- name

- class

- section

- parentId

- admissionDate

Teachers:

- teacherId

- name

- subjects

- contact details

Attendance:

- date
- studentId
- class
- section
- status

TeacherAttendance:

- date
- teacherId
- status

Fees:

- studentId
- totalAmount
- paidAmount
- dueAmount
- paymentHistory

Exams:

- class
- subject
- examDate
- maxMarks

Results:

- studentId

- examId

- marksObtained

Notifications:

- title

- message

- userType

- date

---

7. API Structure

Auth:

- POST /api/auth/login

- POST /api/auth/register

Students:

- POST /api/students/add

- GET /api/students/list

- PUT /api/students/update

- DELETE /api/students/delete

Attendance:

- POST /api/attendance/mark

- GET /api/attendance/report

Fees:

- POST /api/fees/create

- POST /api/fees/pay

- GET /api/fees/receipt

Exams:

- POST /api/exams/create

- POST /api/results/add

---

8. Mobile Application Features

- Secure Login

- Dashboard

- Attendance View

- Homework & Notices

- Exam Results

- Fees Payment

- Push Notifications

- Profile Management

---

9. Security Features

- Role-based access control

- Password encryption (bcrypt)

- JWT authentication

- Input validation

- Secure APIs

---

10. Development Timeline

UI/UX Design: 2 Weeks

Backend Development: 4 Weeks

Admin Panel: 3 Weeks

Mobile App Development: 4 Weeks

Testing & QA: 2 Weeks

Deployment: 1 Week

Total Duration: Approximately 3 Months

---

11. Deliverables

- Admin Web CRM

- Flutter Mobile Applications

- Backend APIs

- Database Setup

- Source Code

- API Documentation

- Deployment Guide

- User Manual

---

12. Future Enhancements

- Face Recognition Attendance

- AI Student Performance Analysis

- Transport GPS Tracking

- Library & Hostel Modules

- Multi-Branch School Support

End of Document

Cripcocode Artificial Intelligence Pvt. Ltd.

Director

Cripcocode Artificial Intelligence Pvt. Ltd.